

A REPORT
ON
SMARTPRINT AI

Submitted by

BRAHMI- 20241CSE1078

Under the guidance of

Ms.Saiqa Khan N

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

AT



PRESIDENCY UNIVERSITY

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PRESIDENCY UNIVERSITY

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING



CERTIFICATE

This is to certify that the report of **CSE7000 – Internship** on “**AI Centre of Excellence (Accelerated by NVIDIA)**” being submitted by “**BRAHMI**” bearing roll number “**20241CSE1078**” in partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology in Computer Science and Engineering (Computer Science and Engineering)** is a bonafide work carried out under our supervision.

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DECLARATION

I hereby declare that the work, which is being presented in the report entitled “**AI Centre of Excellence (Accelerated by NVIDIA)**” in partial fulfillment for the award of Degree of Bachelor of Technology in **Computer Science and Engineering (Computer Science and Engineering)**, is a record of my own investigations carried under the guidance of **Ms.Saiqa Khan N, Assistant Professor and Ms.Delsy Jyothi, Assistant Professor**, Presidency School of Computer Science and Engineering, Presidency University, Bengaluru.

I have not submitted the matter presented in this report anywhere for the award of any other Degree.

BRAHMI (20241CSE1078)

INTERNSHIP COMPLETION CERTIFICATE

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ABSTRACT

SmartPrintAI-An Intelligent AI-Based Printing Queue Optimization System

The rapid growth of digital learning has significantly increased the demand for printing services in educational institutions. In universities and colleges, students frequently print assignments, laboratory records, project reports, examination documents, research papers, and various academic materials. During short class breaks or before assignment submission deadlines, printing centers experience heavy congestion, resulting in long queues, extended waiting times, manual document handling, and inefficient printer utilization. Traditional printing systems largely depend on face-to-face interactions, where students must physically visit the printing shop, wait for their turn, transfer files using USB drives or messaging applications, communicate printing preferences verbally, and remain at the shop until printing is completed. These conventional methods consume valuable time, create confusion during peak hours, increase the workload of printing operators, and often lead to document mix-ups or incorrect print settings. Such challenges reduce the overall efficiency of campus printing services and negatively affect the academic experience of students.

To address these issues, this internship project presents **SmartPrintAI**, an intelligent AI-based campus printing management and queue optimization system. SmartPrintAI is designed to automate the complete printing workflow while minimizing manual intervention. The system allows students to upload documents through a web application from anywhere within the campus, select printing preferences such as color or black-and-white printing, single-sided or double-sided printing, page range selection, number of copies, paper size, and priority printing options. After selecting the desired print configuration, users can complete secure online payments through an integrated payment gateway. Once payment is confirmed, the document is automatically placed into a smart print queue based on various scheduling parameters, eliminating the need for students to stand in long physical queues.

A key feature of SmartPrintAI is the integration of **Artificial Intelligence (AI)** and **Deep Learning** techniques to enhance decision-making and optimize printing operations. The system employs AI algorithms to estimate document processing time, predict queue waiting time, analyze printer workload, and recommend the most suitable printer based on its availability and current status. Deep learning models, particularly Convolutional Neural Networks (CNNs), can be utilized to evaluate uploaded document quality, detect low-resolution images, identify blurred or damaged pages, and provide recommendations before printing begins. This intelligent preprocessing minimizes paper wastage and reduces the occurrence of reprinting due to poor document quality. Furthermore, the queue optimization algorithm dynamically schedules print jobs by considering factors such as document size, printer speed, queue length, and priority requests, thereby improving resource utilization and reducing average waiting time.

The proposed system consists of several interconnected modules, including user

authentication, document upload, printer selection, AI-based queue prediction, payment processing, automatic print scheduling, notification services, and administrative monitoring. Students receive a unique print identification number after successful payment, enabling them to collect printed documents without confusion. Real-time notifications keep users informed about queue status, estimated completion time, and document readiness. The administrative interface enables printing staff to monitor printer health, manage connected printers, observe queue statistics, and analyze system performance through dashboards without manually organizing print jobs.

SmartPrintAI is developed using modern web technologies with a scalable architecture. The frontend provides an intuitive and user-friendly interface, while the backend manages document processing, payment verification, printer communication, and database operations. Technologies such as Python, TensorFlow, OpenCV, SQLite/MySQL, REST APIs, and NVIDIA GPU acceleration can be integrated to support AI model training and inference efficiently. GPU-based processing significantly reduces computation time for deep learning tasks and enables faster document quality analysis.

The proposed solution offers several advantages over traditional campus printing systems. It substantially reduces physical queues, minimizes waiting time, decreases manual errors, optimizes printer utilization, improves user convenience, supports digital payments, and provides a transparent printing process through real-time tracking. Additionally, priority printing services can assist students facing urgent submission deadlines while maintaining fairness through intelligent scheduling algorithms.

In conclusion, **SmartPrintAI** demonstrates how Artificial Intelligence, Deep Learning, and modern web technologies can be combined to solve a real-world campus problem effectively. By automating the printing workflow and intelligently managing print queues, the system enhances operational efficiency, improves student satisfaction, and supports digital transformation within educational institutions. The project also provides a strong foundation for future enhancements, including multi-campus deployment, cloud-based print management, mobile application integration, IoT-enabled smart printers, predictive maintenance, voice-assisted printing, and advanced analytics for resource optimization. Overall, SmartPrintAI represents an innovative, practical, and scalable solution that aligns with the objectives of Industry 4.0 and intelligent campus infrastructure, making it a valuable contribution to modern educational technology.

The development of SmartPrintAI also provided practical exposure to the complete AI application development lifecycle, including problem identification, system design, data and document processing, AI model integration, testing, debugging, and performance evaluation. The modular design of the system allows individual components such as AI-based document analysis, queue prediction, payment processing, and printer management to be improved or extended independently in future implementations.

INTRODUCTION

The rapid advancement of **Information Technology** and **Artificial Intelligence (AI)** has transformed the way educational institutions manage academic and administrative activities. Most university services such as course registration, attendance management, online examinations, and digital learning have become automated through web-based applications. However, campus printing services in many educational institutions still rely on traditional manual methods, requiring students to visit printing centers physically, wait in long queues, and interact directly with printing operators. This conventional approach often leads to delays, overcrowding, and inefficient utilization of printing resources.

During assignment submission deadlines, laboratory sessions, project demonstrations, and examination periods, printing centers experience heavy workloads due to the large number of students requesting printing services simultaneously. Since print requests are handled manually, students spend a significant amount of time waiting for their turn, while printing operators must manage document collection, print configuration, payment processing, and printer allocation manually. These challenges reduce operational efficiency and negatively affect the overall user experience.

The increasing demand for faster and more convenient printing services has created a need for an intelligent printing management solution capable of automating the complete printing workflow. A modern printing platform should allow students to upload documents remotely, configure printing preferences, complete secure online payments, monitor print status, and collect printed documents without unnecessary waiting. Such a system should also optimize printer utilization and reduce the workload of printing operators.

To address these challenges, this project proposes **SmartPrintAI**, an **AI-powered intelligent printing management system** developed using modern web technologies and Artificial Intelligence techniques. The system enables students to upload documents online, select printing options, perform secure digital payments, and track print requests in real time. Unlike traditional printing systems, SmartPrintAI incorporates intelligent queue management and automated printer allocation to improve printing efficiency and reduce waiting time.

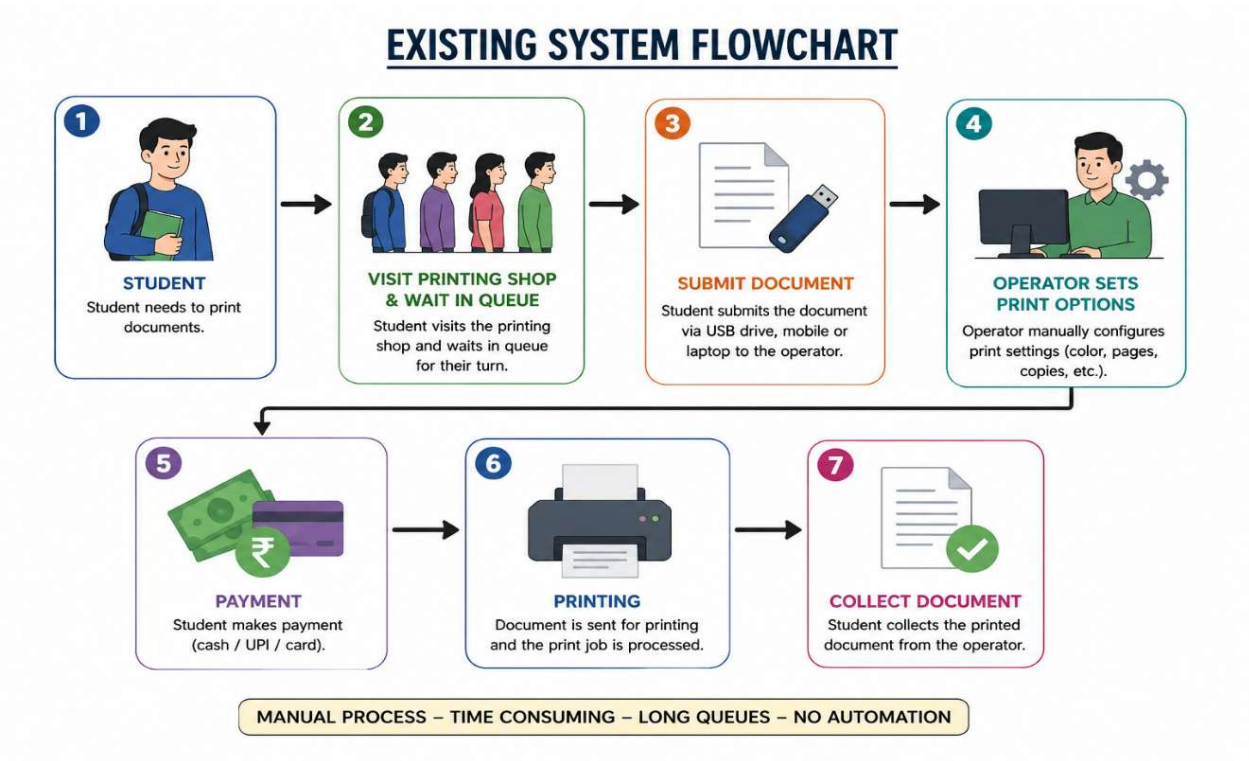
One of the major features of **SmartPrintAI** is its **AI-based document analysis**. Before sending a document for printing, the system analyzes its quality using **Deep Learning** and **Computer Vision** techniques. The AI engine checks for image clarity, resolution, orientation, and formatting issues, helping users identify potential printing problems before the document is processed. This minimizes printing errors, reduces paper wastage, and improves print quality.

The system also provides **real-time queue management**, allowing users to monitor their queue position and estimated waiting time. Once the payment is completed, the print request is automatically scheduled and assigned to the most suitable printer based on its availability and workload. A unique **Print ID** is generated for every print request, ensuring secure and error-free document collection.

Overall, **SmartPrintAI** demonstrates how **Artificial Intelligence**, **Deep Learning**, and modern web technologies can be integrated to improve campus printing services. The proposed solution provides a faster, smarter, and more efficient printing experience while reducing manual intervention and supporting the digital transformation of educational institutions.

1.1 Existing System

The existing printing system used in many educational institutions follows a **manual workflow**. Students must physically visit the printing center, wait in long queues, submit their documents to the printing operator, select printing preferences, make payments manually, and wait until the printing process is completed. During peak academic hours, this process becomes time-consuming and inconvenient, resulting in long waiting times and poor printer utilization.

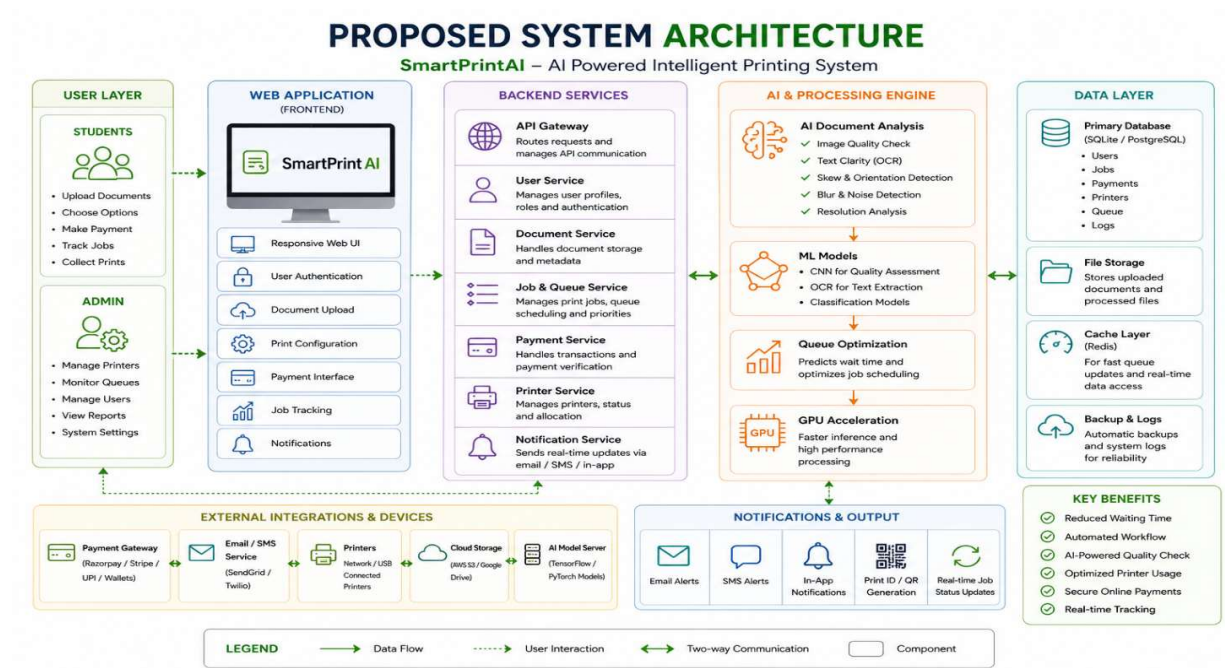


1.2 Problem Statement

The traditional campus printing process suffers from several limitations, including manual document submission, long waiting queues, offline payment methods, lack of queue tracking, and inefficient printer allocation. Students often waste valuable academic time waiting for print services, while printing operators face difficulties managing multiple print requests simultaneously. These issues highlight the need for an intelligent and automated printing management solution.

1.3 Proposed System

The proposed SmartPrintAI system provides a web-based platform that enables students to upload documents remotely, configure print settings, perform secure online payments, and monitor their print status in real time. The system uses Artificial Intelligence (AI) to analyze document quality, optimize print queues, and allocate printers automatically. This approach reduces waiting time, improves printer utilization, minimizes human errors, and enhances the overall printing experience.



OBJECTIVE

The primary objective of **SmartPrintAI** is to develop an intelligent, AI-powered campus printing management system that simplifies the printing process by reducing waiting time, automating print queue management, and improving the overall user experience. The system aims to provide students with a fast, secure, and convenient platform for submitting print requests while enabling administrators to efficiently monitor and manage printing resources.

2.1 Main Objective

The main objective of this project is to design and implement a smart web-based printing system that integrates Artificial Intelligence, Machine Learning, and modern web technologies to automate the complete printing workflow, from document upload to final print collection.

2.2 Specific Objectives

- To develop a secure web application for students and administrators.
- To enable online document upload from any location within the campus.
- To allow users to select printing preferences such as page size, color mode, orientation, and number of copies.
- To integrate AI-based document analysis for checking document quality before printing.
- To implement an intelligent queue management system that predicts waiting time and optimizes print scheduling.
- To provide secure online payment for print services.
- To automate printer allocation based on printer availability and workload.
- To display real-time queue status and printing progress to users.
- To generate a unique Print ID for secure document collection.
- To reduce paper wastage and minimize printing errors through AI-assisted processing.
- To improve operational efficiency by reducing manual intervention.
- To provide an easy-to-use, responsive, and user-friendly interface for both students and administrators.

2.3 Expected Outcomes

The implementation of SmartPrintAI is expected to achieve the following outcomes:

- Reduced waiting time at campus printing centers.
- Faster and more efficient print job processing.
- Improved utilization of available printers.
- Enhanced document quality through AI-based analysis.

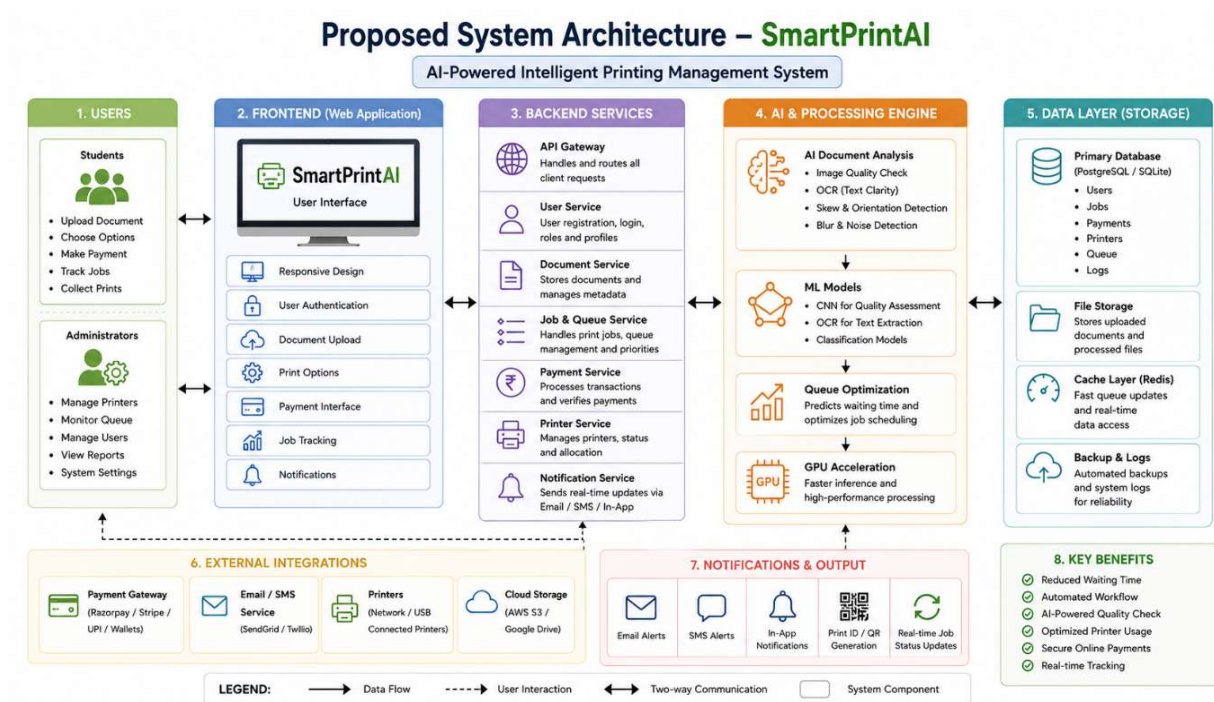
- Transparent and real-time print queue tracking.
- Secure and cashless payment process.
- Better user experience for students and printing administrators.
- A scalable and intelligent printing solution suitable for educational institutions.

WORK PROGRESS

3.1 System Overview

The SmartPrintAI project was developed to automate and modernize the campus printing process using Artificial Intelligence (AI) and modern web technologies. The system provides an integrated platform where students can upload documents, select print settings, make secure online payments, track print requests, and collect printed documents efficiently. The implementation follows a modular architecture, allowing each component to perform a dedicated function while ensuring smooth communication between all modules.

The project consists of two primary interfaces: the Student Portal and the Administrator Portal. The Student Portal enables users to upload documents, monitor print queues, and manage print requests, while the Administrator Portal provides facilities to monitor printers, manage users, view reports, and supervise overall system performance.

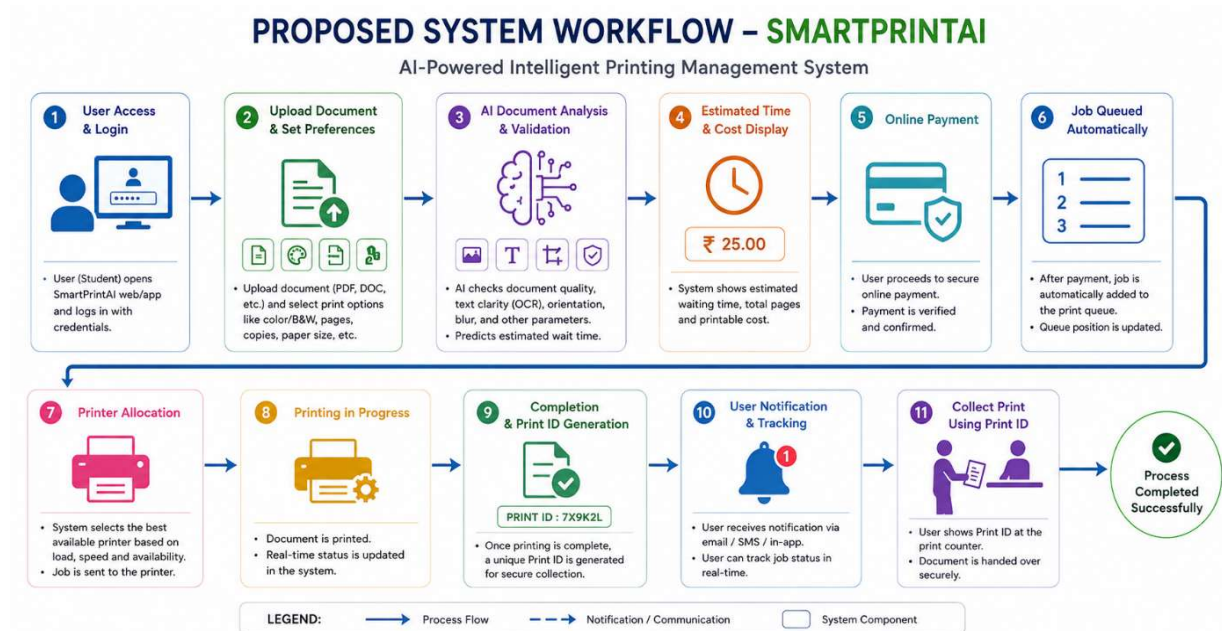


3.2 Project Workflow

The SmartPrintAI workflow begins when a student accesses the web application and logs into the system using valid credentials. After successful authentication, the user uploads a document for printing and selects the required printing preferences. The uploaded document is analyzed by the AI engine, which evaluates document quality and predicts the estimated waiting time.

Once the analysis is completed, the user proceeds with online payment. After successful

payment verification, the system automatically schedules the print job, allocates an available printer, and updates the queue status. Throughout the printing process, the user receives real-time notifications and can monitor the progress through the dashboard. After printing is completed, the generated Print ID is used to collect the document securely.



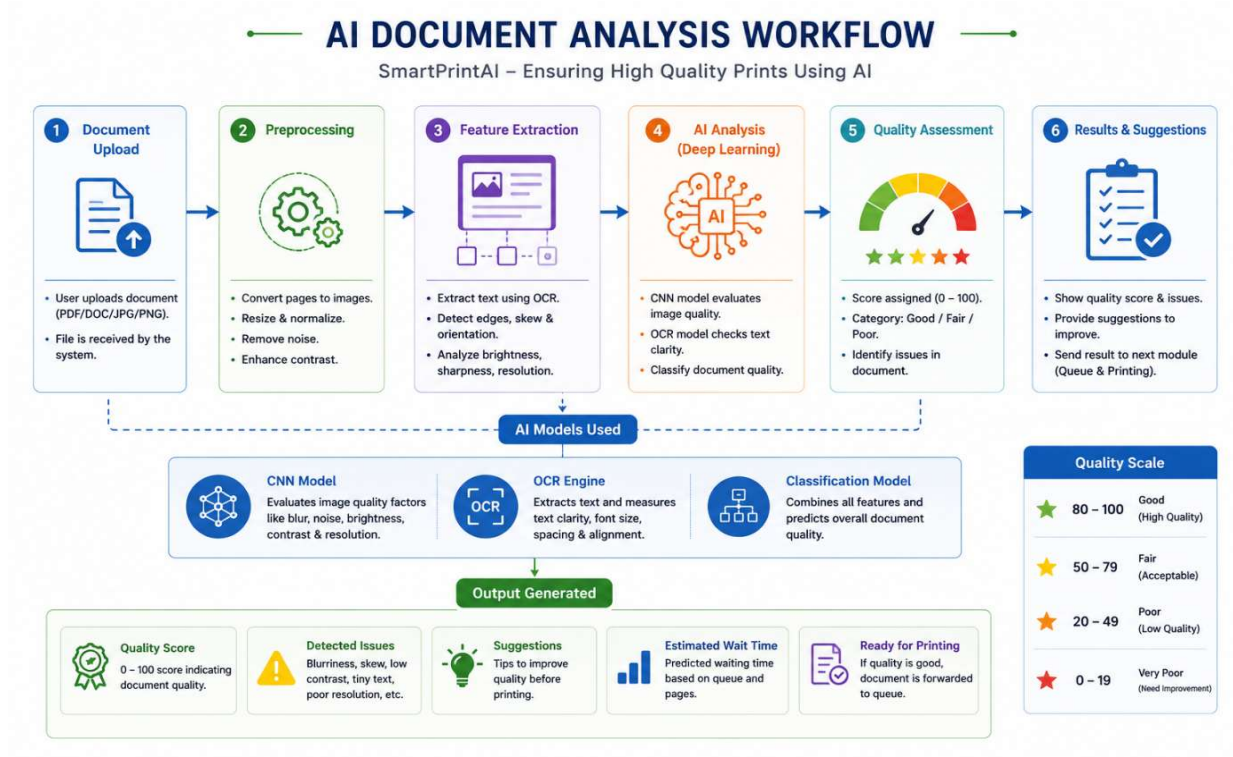
3.3 Technologies Used

The SmartPrintAI system is implemented using the following technologies:

Technology	Purpose
React.js	Frontend development
TypeScript	Application logic
Tailwind CSS	Responsive UI design
Python	AI and backend processing
TensorFlow	Deep Learning model
Technology	Purpose
OpenCV	Document image analysis
SQLite / PostgreSQL	Database management
FastAPI	Backend API services
NVIDIA GPU (CUDA)	AI model acceleration

3.4 AI-Based Document Analysis

One of the key features of SmartPrintAI is its AI-powered document analysis module. After a document is uploaded, the AI engine analyzes parameters such as image quality, document orientation, text clarity, and formatting before sending the file for printing. This process helps detect potential issues in advance, reducing printing errors and minimizing paper wastage. The system also predicts the estimated waiting time by considering queue length, printer availability, and document size.



3.5 Database Design

The SmartPrintAI application uses a relational database to store user information, uploaded documents, payment records, queue details, printer information, and system notifications. The database design ensures secure storage, fast retrieval, and efficient management of print requests.

The major database entities include:

- Users
- Documents
- Print Jobs
- Payments
- Printers

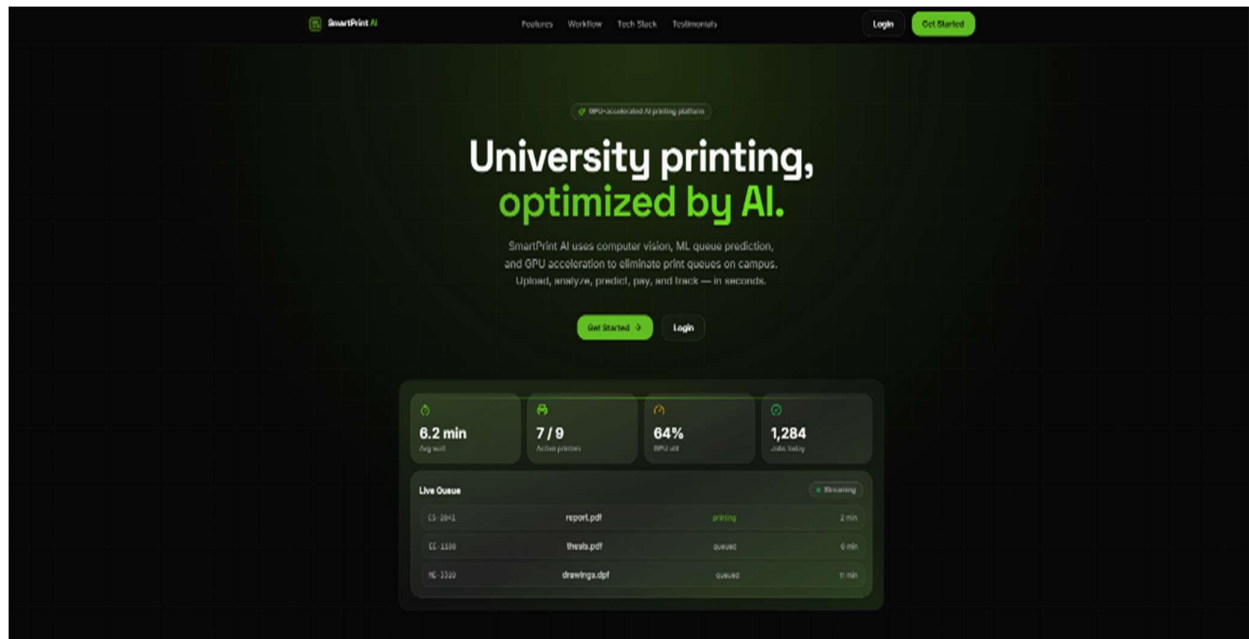
- Queue
- Notifications

3.6 Implementation

The implementation phase involved designing and integrating all modules into a single web application. The project provides separate interfaces for students and administrators to ensure secure access and efficient system management.

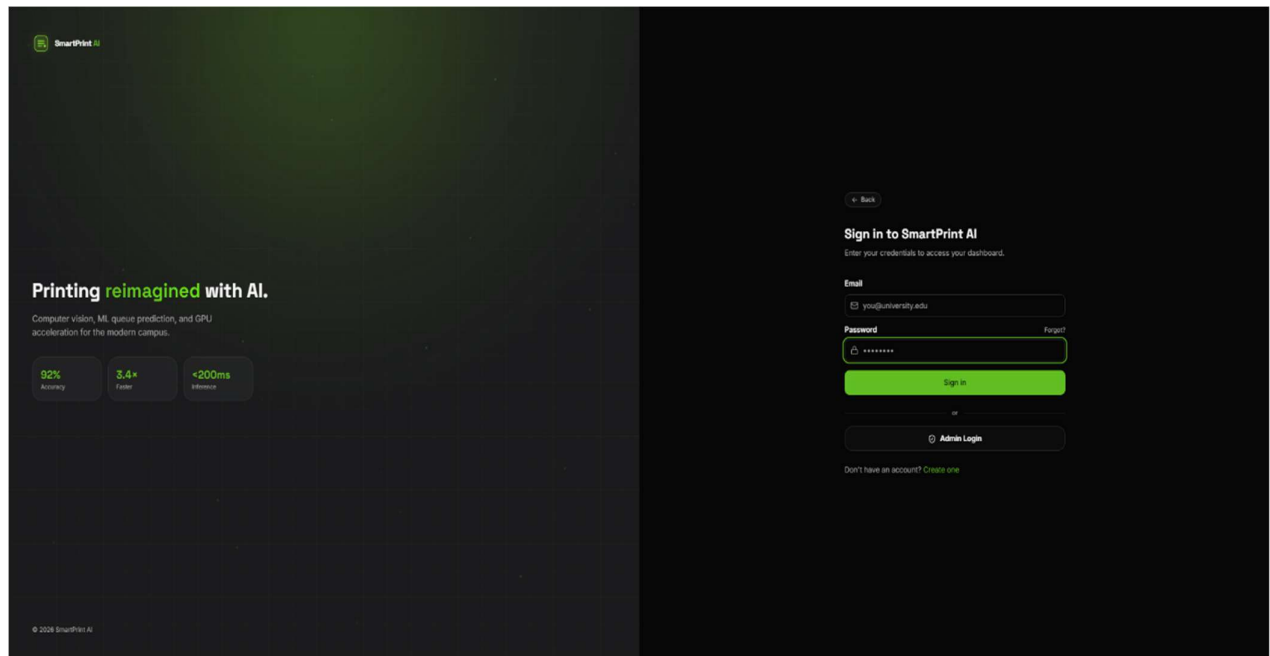
Landing Page

The landing page introduces SmartPrintAI and highlights its AI-powered printing capabilities. It provides navigation to the main features of the application and allows users to register or log in.



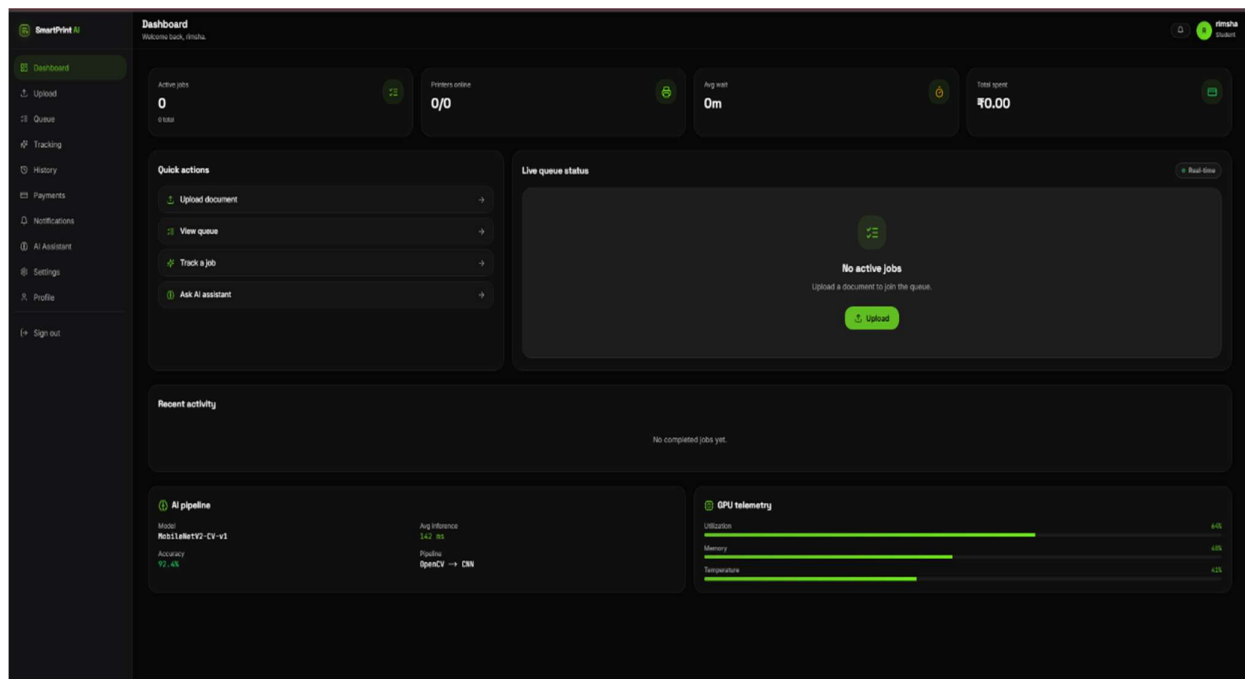
User Authentication

The authentication module enables students and administrators to securely access the system using their registered credentials. Role-based authentication ensures that users can access only the functionalities assigned to them.



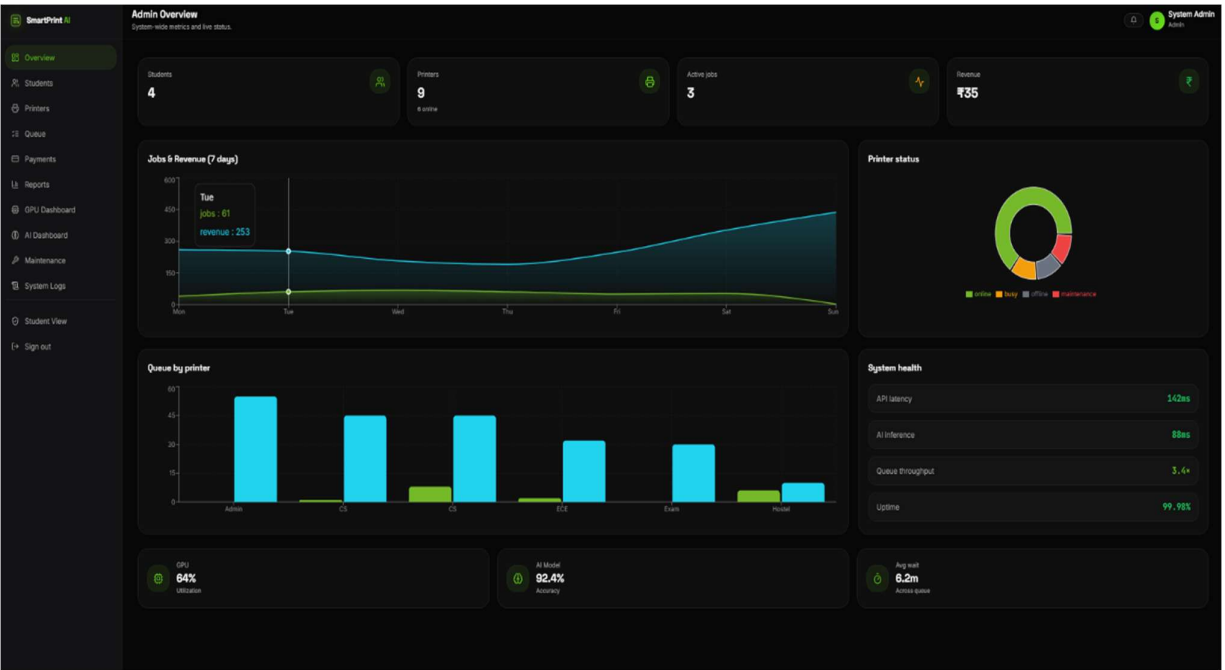
Student Dashboard

The Student Dashboard serves as the central workspace for students. It displays active print jobs, queue status, AI analysis metrics, GPU utilization, recent activity, and quick action buttons for uploading documents and tracking print requests.



Admin Dashboard

The Admin Dashboard enables administrators to monitor printer status, manage users, analyze revenue, view system health, monitor GPU utilization, and generate reports for efficient management of the printing environment.



RESULTS

The SmartPrintAI system was successfully designed and implemented as an intelligent web-based printing management platform. The developed application integrates Artificial Intelligence (AI) with modern web technologies to simplify and automate the campus printing process. The system provides separate interfaces for students and administrators, enabling secure document upload, AI-based document analysis, intelligent queue management, real-time job tracking, and efficient printer management.

The implementation of SmartPrintAI demonstrates that the proposed solution effectively reduces manual intervention and enhances the overall printing experience. Users can upload documents remotely, configure printing options, monitor print status, and complete payments through a single platform. The AI module analyzes uploaded documents before printing, helping improve document quality while minimizing printing errors and paper wastage.

4.1 System Performance

The developed system successfully performs the following operations:

- Secure user authentication for students and administrators.
- Online document upload with multiple file format support.
- AI-based document quality analysis before printing.
- Intelligent queue management and waiting time prediction.
- Real-time print job tracking.
- Automatic printer allocation based on availability.
- Secure online payment processing.
- Administrative monitoring through an interactive dashboard.

4.2 Performance Comparison

Parameter	Existing System	SmartPrintAI
Document Submission	Manual	Online
Queue Management	Manual	AI-Based
Payment Method	Offline	Online
Queue Tracking	Not Available	Real-Time
Printer Allocation	Manual	Automatic
Document Quality Check	Not Available	AI-Based
Waiting Time	High	Reduced
User Experience	Average	Improved
Paper Wastage	Higher	Reduced
System Monitoring	Manual	Dashboard Based

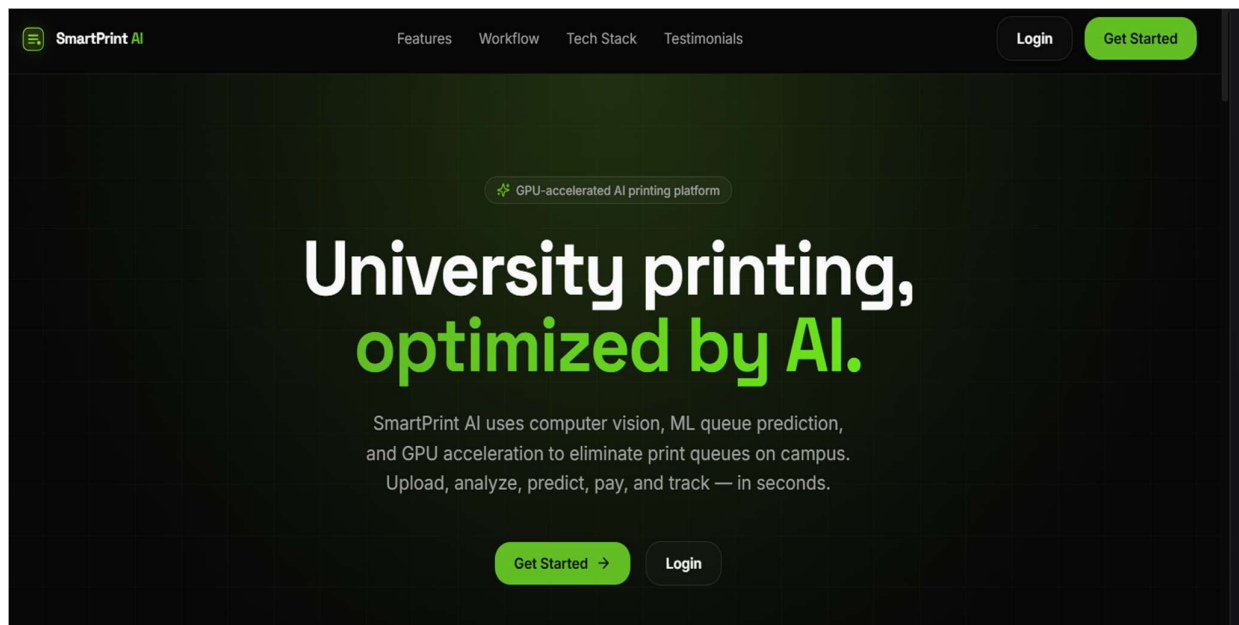
4.3 Advantages Achieved

The implementation of SmartPrintAI provides several improvements over the traditional printing system. The integration of AI enables automatic document quality assessment and intelligent print queue optimization, significantly reducing the waiting time for students. The web-based platform allows users to submit print requests remotely, eliminating the need to stand in long queues.

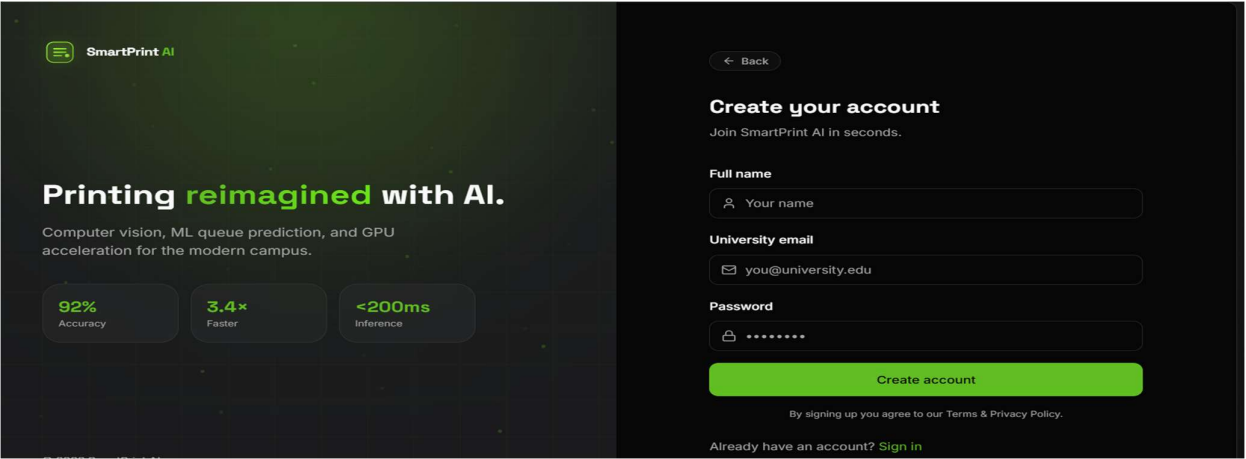
The administrator dashboard simplifies printer management by providing real-time information about printer availability, active jobs, system performance, and revenue statistics. Secure online payment and automated notifications further improve user convenience and system efficiency.

4.4 Overall Outcome

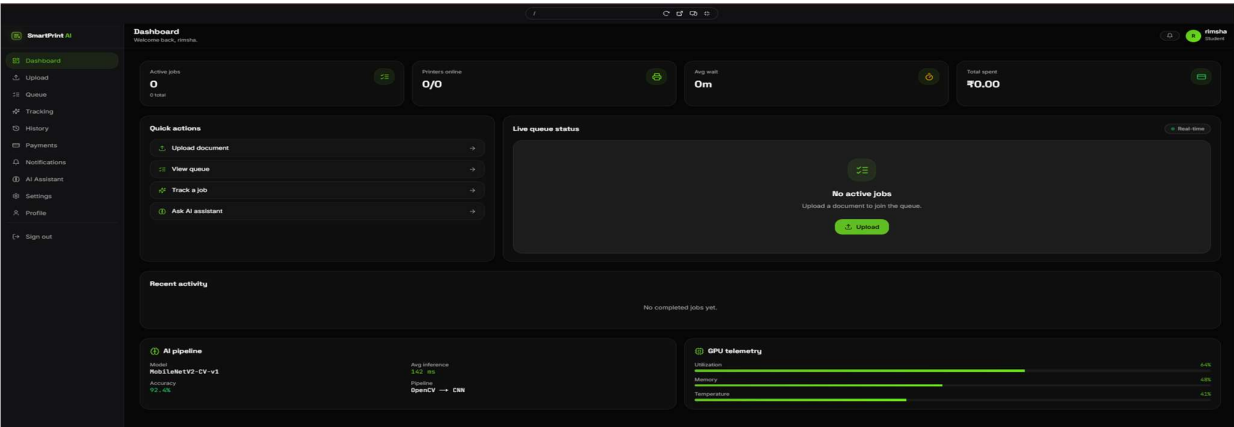
The SmartPrintAI project successfully demonstrates the practical application of **Artificial Intelligence**, **Machine Learning**, and modern web technologies in automating campus printing services. The developed system offers a reliable, user-friendly, and scalable solution that reduces manual effort, improves printer utilization, and enhances the overall efficiency of printing operations. The project meets the intended objectives and provides a strong foundation for future enhancements such as mobile application support, cloud deployment, predictive maintenance, and advanced analytics.



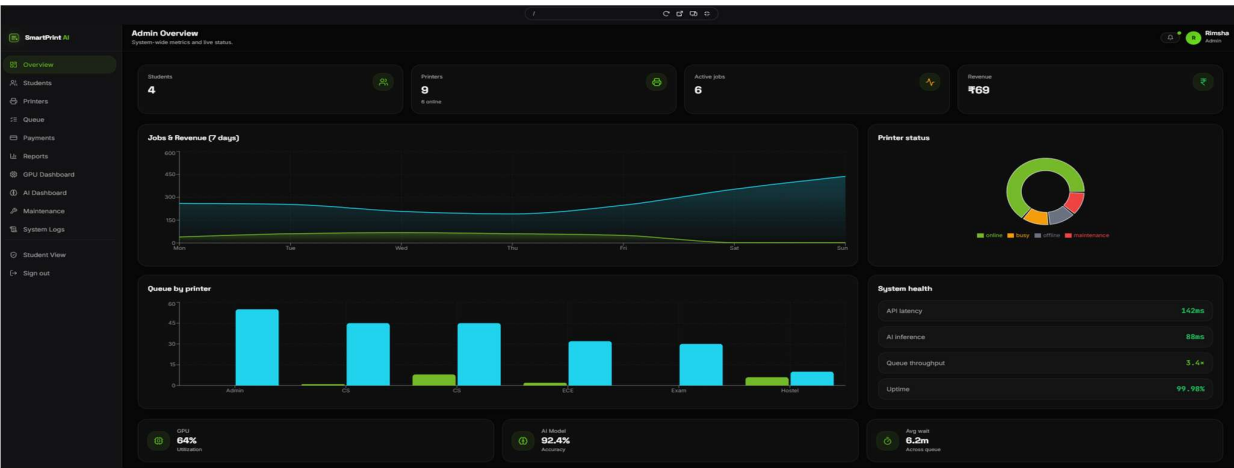
Landing Page



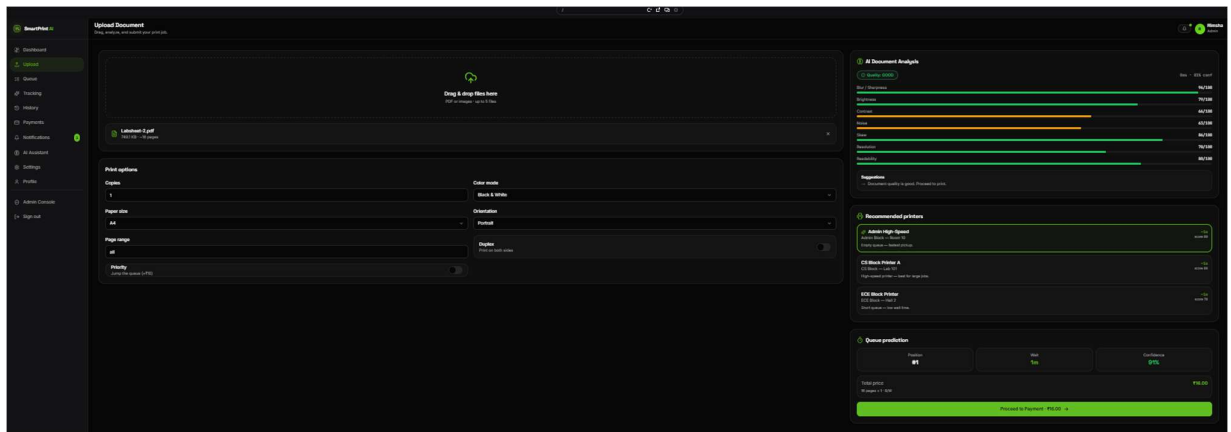
Login Page



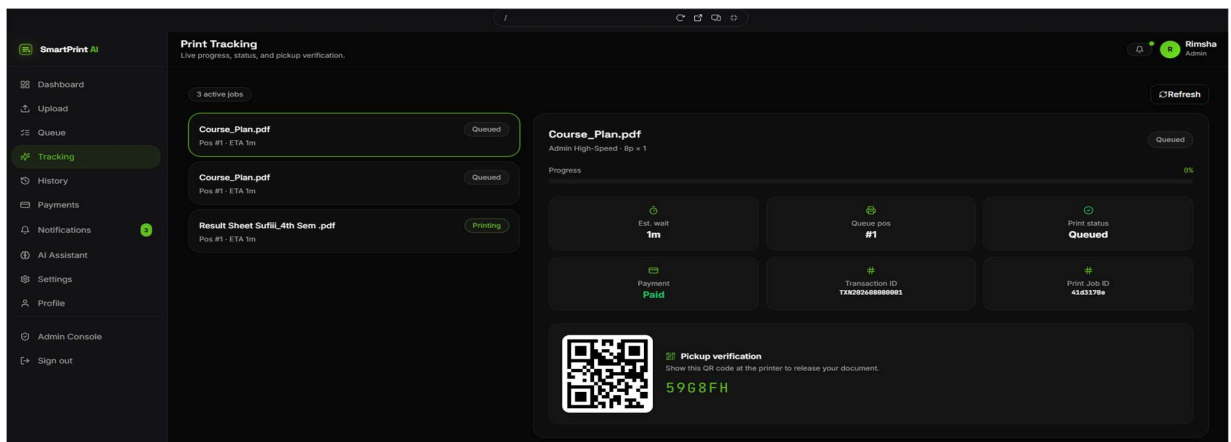
Student Dashboard



Admin Dashboard



Document Upload & Payment



Queue and Tracking



GPU Dashboard in Admin Panel

CONCLUSION

The SmartPrintAI project successfully demonstrates the implementation of an AI-powered intelligent printing management system designed to improve the efficiency of campus printing services. The system replaces the traditional manual printing process with a modern web-based platform that enables students to upload documents, configure printing options, make secure online payments, and monitor print requests in real time.

By integrating Artificial Intelligence, Machine Learning, and Computer Vision, the system performs document quality analysis before printing, helping to identify issues such as low image quality, incorrect orientation, and formatting errors. The intelligent queue management mechanism automatically schedules print jobs and allocates available printers, reducing waiting time and improving printer utilization.

The developed application provides separate dashboards for students and administrators, making the system user-friendly and easy to manage. Features such as online document submission, AI-assisted document analysis, real-time queue tracking, secure payment processing, and automated printer allocation contribute to a more reliable and efficient printing environment.

Overall, the objectives of the project have been successfully achieved. SmartPrintAI demonstrates how Artificial Intelligence and modern web technologies can be applied to solve a real-world problem faced by educational institutions. The project provides a scalable, secure, and practical solution that enhances the overall printing experience while reducing manual effort and operational delays.

Future Scope

Although the current implementation successfully achieves its objectives, several enhancements can be incorporated in future versions to further improve the system's capabilities:

- Develop a dedicated mobile application for Android and iOS platforms.
- Integrate cloud-based storage for secure document management.
- Support multiple campus printing centers through a centralized platform.
- Implement AI-based predictive maintenance for printers to reduce downtime.
- Introduce QR code-based document collection for faster and more secure retrieval.
- Add voice-assisted printing using AI-based virtual assistants.
- Incorporate advanced analytics and reporting for monitoring printer usage and system performance.
- Expand the system to support commercial printing services beyond educational institutions.

Overall Summary

The SmartPrintAI project demonstrates the successful integration of Artificial Intelligence, Deep Learning, and modern web technologies to automate and optimize campus printing services. By reducing waiting time, improving document quality, automating queue management, and enhancing user convenience, the system provides an innovative solution that addresses the limitations of traditional printing methods. The project has strong potential for further development and practical deployment in educational institutions and other organizations requiring efficient printing management.

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